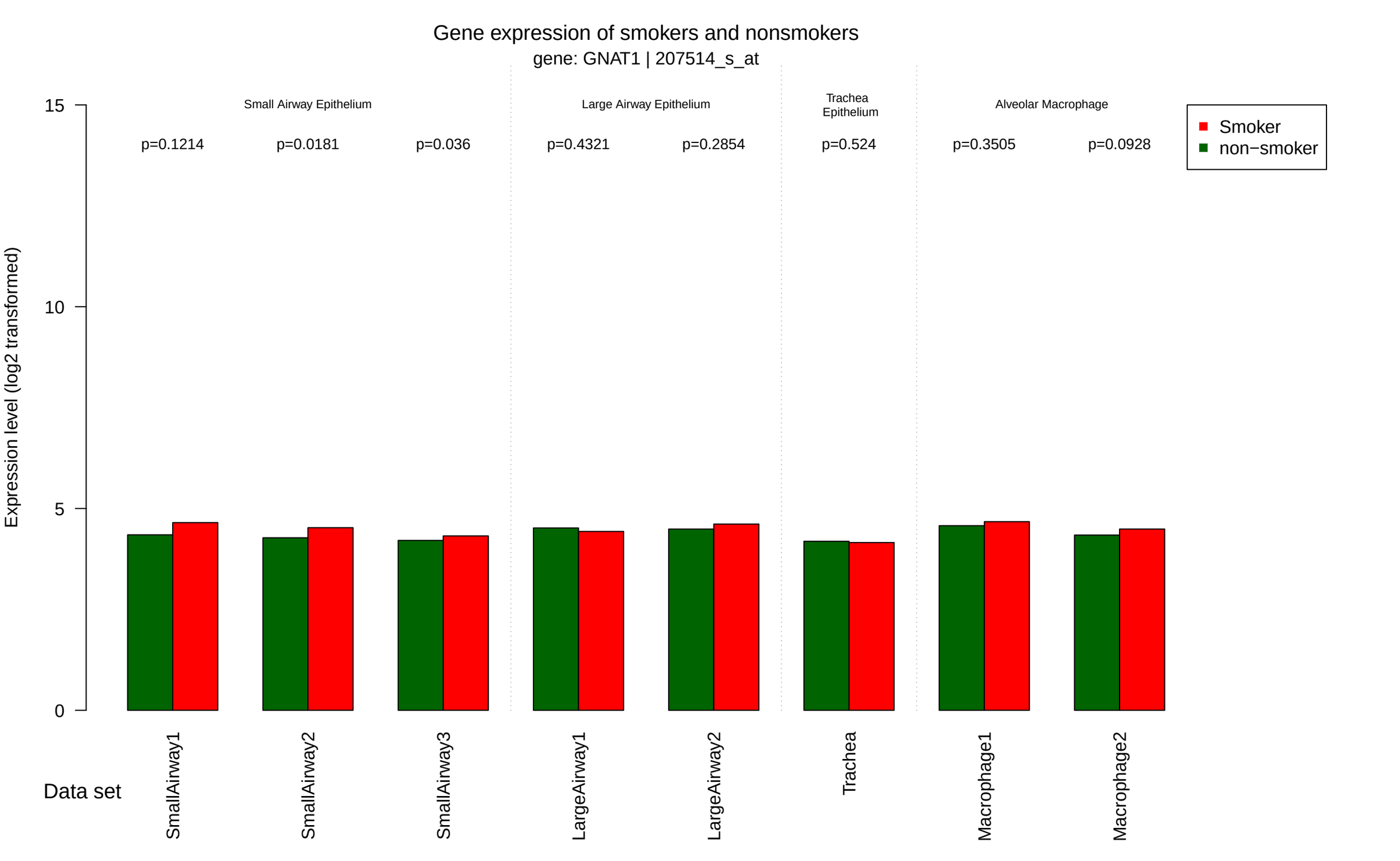
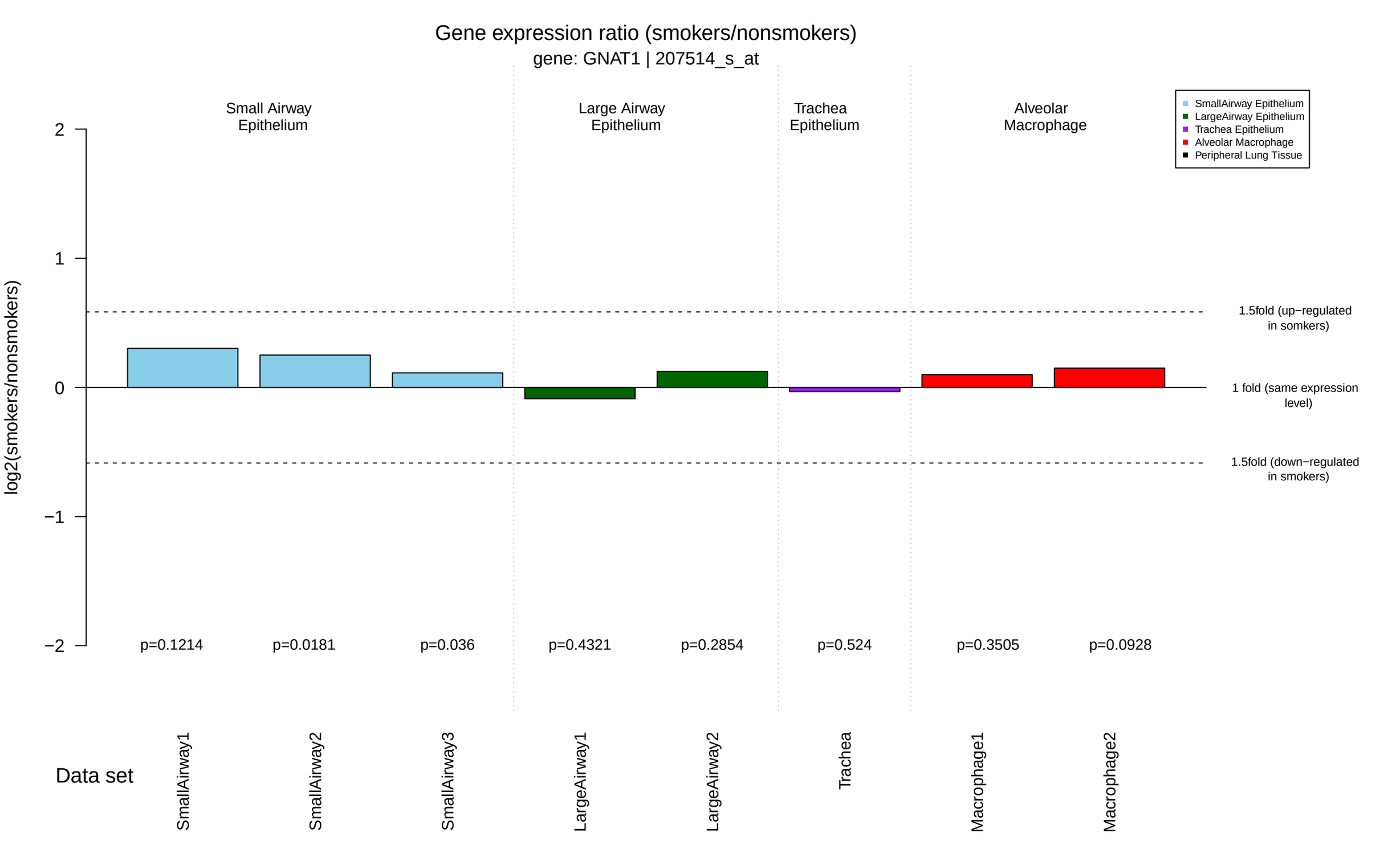
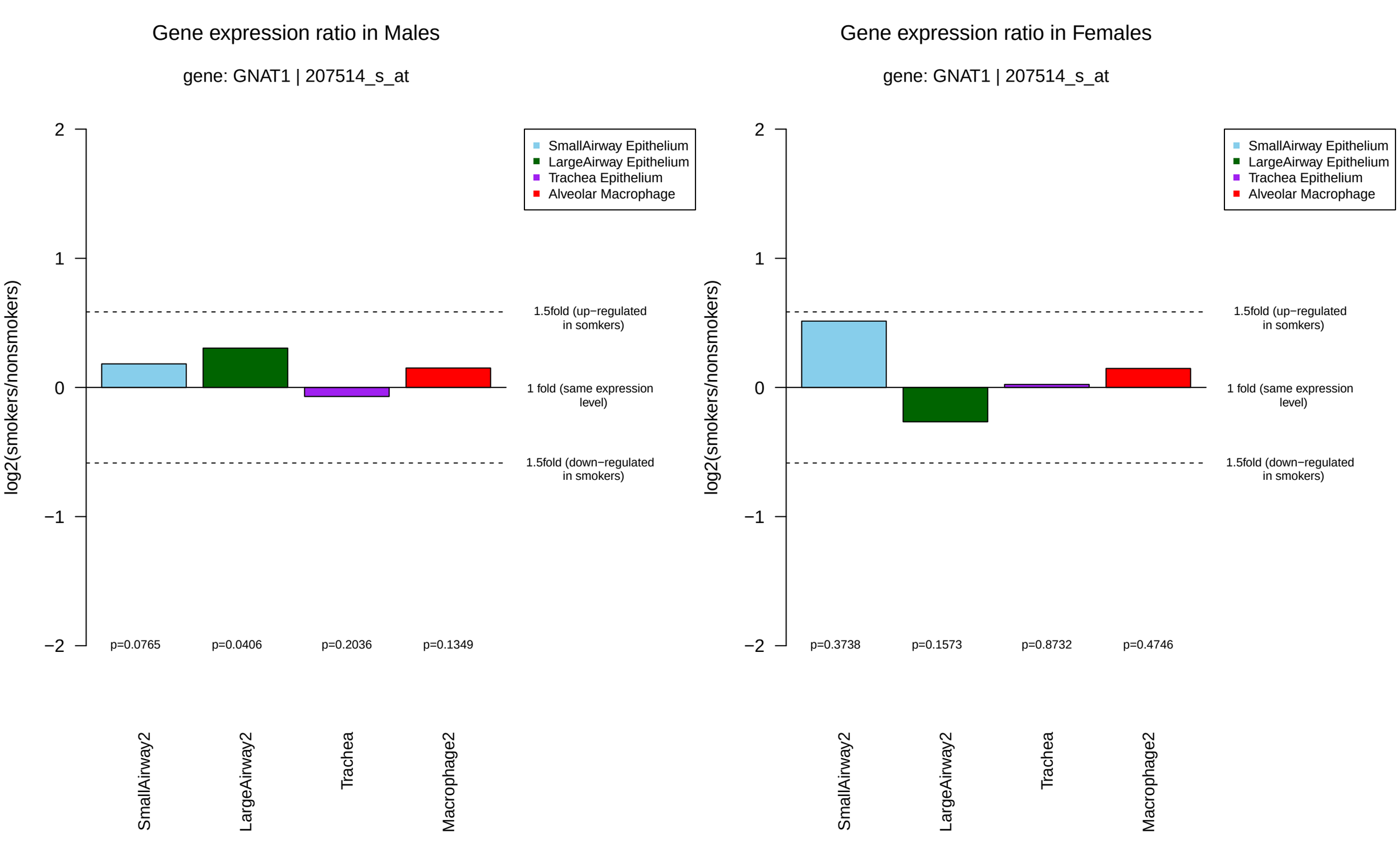
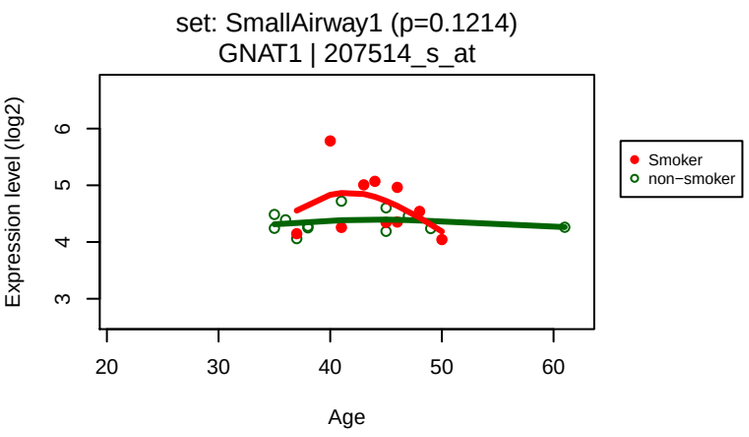
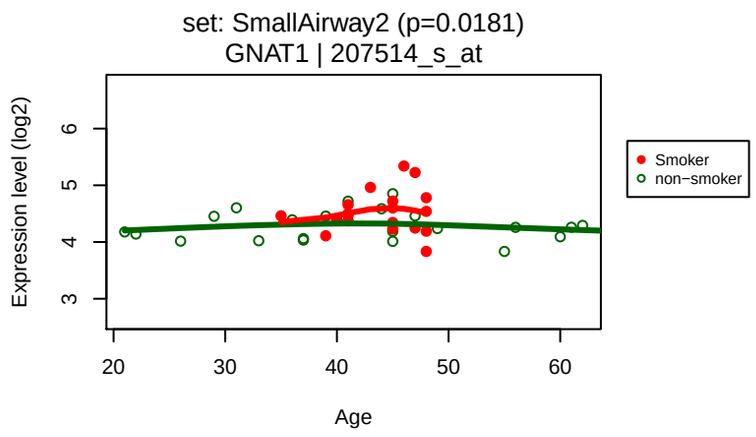
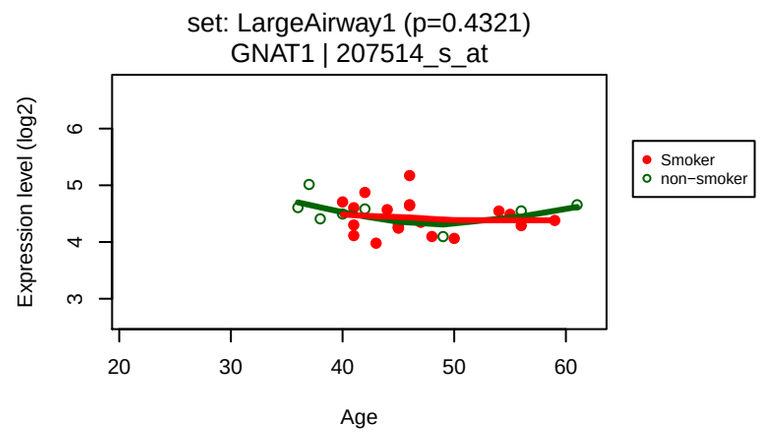

Gene expression of smokers and nonsmokers
gene: GNAT1 | 207514_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1214
p=0.0181
p=0.036
p=0.4321
p=0.2854
p=0.524
p=0.3505
p=0.0928
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

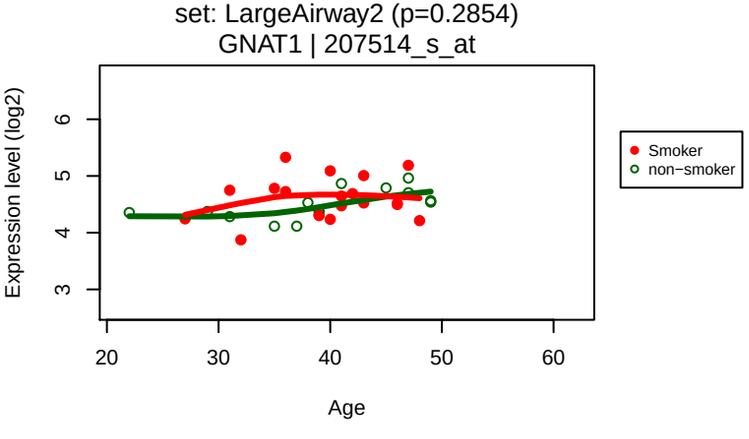

Gene expression ratio (smokers/nonsmokers)
gene: GNAT1 | 207514_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1214
p=0.0181
p=0.036
p=0.4321
p=0.2854
p=0.524
p=0.3505
p=0.0928
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

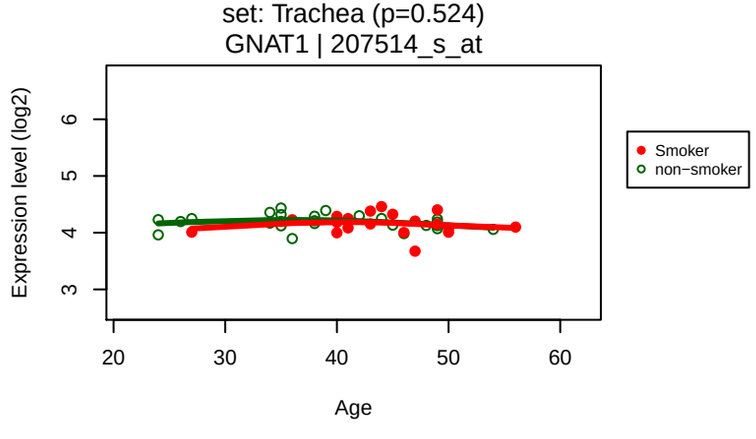

Gene expression ratio in Males
gene: GNAT1 | 207514_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0765
p=0.0406
p=0.2036
p=0.1349
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAT1 | 207514_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3738
p=0.1573
p=0.8732
p=0.4746
SmallAirway2
LargeAirway2
Trachea
Macrophage2

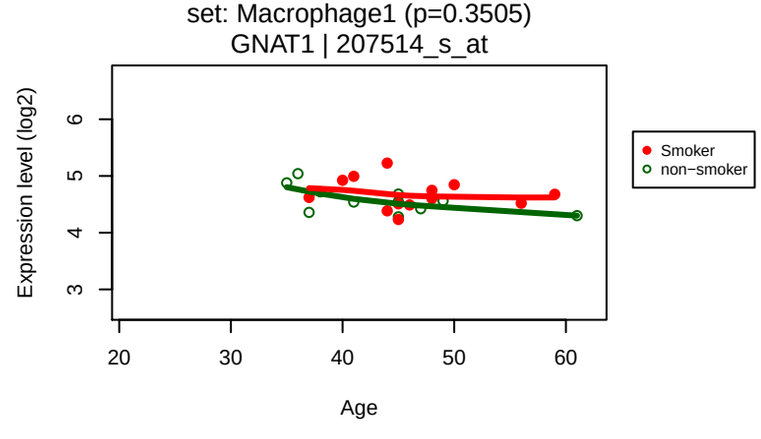

set: SmallAirway1 (p=0.1214)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker

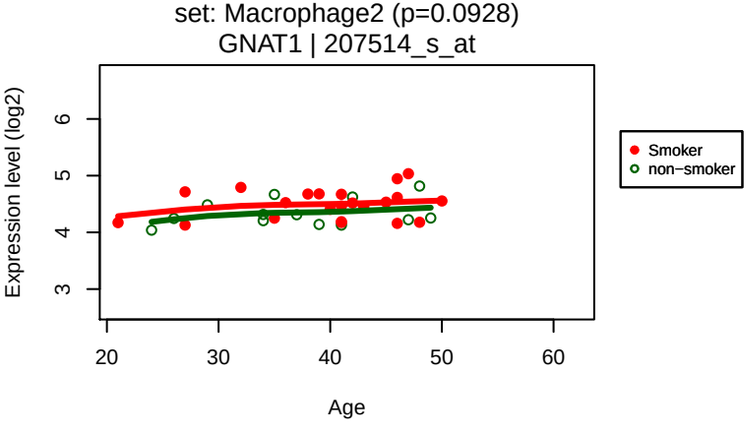

set: SmallAirway2 (p=0.0181)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker

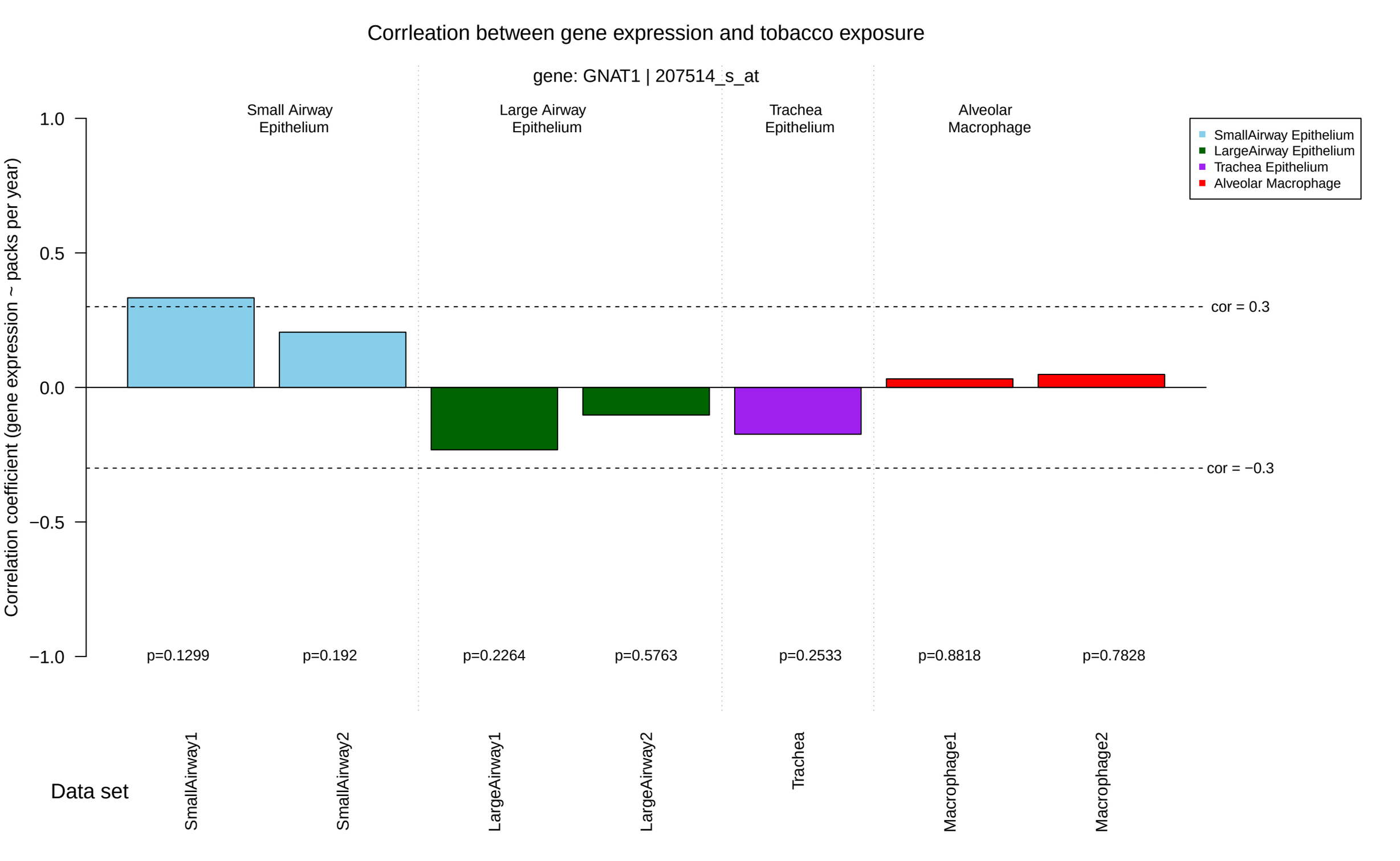

set: LargeAirway1 (p=0.4321)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2854)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.524)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3505)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0928)
GNAT1 | 207514_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAT1 | 207514_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1299
p=0.192
p=0.2264
p=0.5763
p=0.2533
p=0.8818
p=0.7828
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2